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Energy, Economy and Environmental Assessment of E-Tricycle for Medium-Size Cargo Deliveries: A Case of Dar es Salaam City

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ABSTRACT

As global cities become urbanised, cargo deliveries emerge to be crucial for last-mile delivery of daily needs and economic vitality. In Sub-Sahara Africa, especially Tanzania, there is a spike in the adoption of three-wheelers, particularly electric ones, in last-mile delivery. However, their sustainability concerns in cargo deliveries remain underexplored. This study aims to evaluate the performance of the e-tricycle in terms of energy, economy, and environment when deployed for medium-size cargo delivery in targeted routes of Dar es Salaam city, Tanzania. The performance of e-tricycle was compared with those of traditional delivery modes such as petrol-powered motorcycles, tuk-tuk, and light-duty car (LDC). The e-tricycle was revealed to consume up to 0.54 kWh/one-way trip and 1.07 kWh/km while the motorcycle, tuk-tuk, and LDC consumed approximately 18.54, 35.60, and 44.50 kWh/km, depending on the route length. Economically, e-tricycle, motorcycle, tuk-tuk, and LDC cost around TZS 120, 220, 320, and 870 per km while the cost per trip reached TZS 2950, 5200, 3250, and 15,700. When restricting to the delivery weight of 250 kg, the cost per kg for the e-tricycle, motorcycle, tuk-tuk, and LDC could reach around TZS 50, 110, 150, and 380. Replacing the motorcycle, tuk-tuk, and LDC with an e-tricycle could reduce between 10.81 and 43.24 kgCO₂ per day, or between 2.70 and 10.81 kgCO₂ per trip. The insights from this study could be used to improve future city planning by integrating e-tricycle delivery as a low-cost and clean mode of transportation.

1 | Introduction

The rapid urbanisation and population growth [1, 2] in the globe have spiked the need for public transportation [3, 4]. As more than half of the world's population now lives in cities [1, 4], road freight has become the most popular mode [5] due to its low cost, rapidity, and scheduled delivery. Aside from passenger mobility,

urban goods delivery and pickup are essential components of urban living [6]. In other words, urban delivery plays a crucial role in maintaining the economic vibrancy of cities because well-managed urban logistics translate into lower-cost products for consumers and more competitive manufacturing for companies [7]. However, urban mobility accounts for a significant amount of urban daily traffic, causing pollution, noise, and safety concerns.

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Light goods vehicles, such as carrier vans, are responsible for a significant portion of the cost, congestion, and emissions in cities [8].

In urban areas, last-mile deliveries are imperative due to the high density of activities and people [9]. Furthermore, the last mile — which is marked by considerable delays and expensive delivery — often poses a challenge to urban freight management. For instance, in New York City, more than 1.5 million cargos are delivered daily, resulting in congestion and air pollution [10]. In Thailand, the food and restaurant industries, which account for 23% of the gross domestic product (GDP) in the Thai economy, have approximately 116,423 street vendors and food vendors and more than 3 million motorcyclists transporting food in Bangkok alone (2019) [11]. Amidst the pressing issues of climate change and mounting worries about energy scarcity, the government and freight industry players have been actively investigating ways to cut down energy consumption and carbon dioxide (CO₂) emissions from inner-city delivery routes [12]. In the growing concerns over urban air pollution, world megacities have introduced electric vehicles (EVs) [13], especially electric tricycles (e-tricycles), owing to no tailpipe emissions during driving, less noise [14] and smooth driving. The transition to e-tricycles for urban delivery is becoming undeniably necessary. As megacities expand and the demand for efficient, eco-friendly delivery solutions rises [15], e-tricycles emerge as the spine of urban deliveries. Thus, the widespread adoption of e-tricycles paves the way for a sustainable and efficient urban delivery system. However, for the e-tricycle to successfully perform urban delivery, knowledge is needed regarding their operational performance.

A few years ago, efforts were made to understand the performance of the e-tricycles. Balaria et al. (2016) analysed the sustainability of e-tricycles adoption in Cabanatuan City, Philippines. Their sustainability targeted economic, socio-political, and environmental aspects. They revealed that the sustainability of e-tricycle could be viable in the long run. Nacion, Dimla, and Soriano [16] examined the operating efficiency of e-tricycles versus the traditional motorised tricycle. The findings showed that the e-tricycles had the shortest mean wait time. Additionally, according to the commuter-citizen use rates at both terminals, weekday time spent at the motorised terminal is 1.488 min, while at the e-tricycle terminal, it is 1.338 min. On weekends, however, the commuter-citizen spent 16.2 seconds at the e-trike terminal and 18.2 seconds at the motorised terminal, proving that the e-trike terminal is a faster place for commuters and people to wait than the motorised terminus. Zazueta-Nassif et al. [17] did a study on the need to promote the use of more environmentally friendly distribution vehicles in last-mile operations by transitioning away from diesel vans. A survey was conducted by the Colombian research firm Logyca, headquartered in Bogotá. The findings revealed that replacing conventional trucks with a combination of e-bikes and e-tricycles saves around 50% of the cost. Smith and Johnson (2020) examined the impact of e-tricycles on urban air quality. The study utilises air quality monitoring data from several cities and compares the emissions of e-tricycles with traditional delivery vehicles. The findings indicate that e-tricycles significantly reduce air pollution levels in urban areas. The study also highlights the importance of promoting the adoption of e-tricycles for urban delivery to improve air quality and mitigate

the negative effects of vehicle emissions. In General Santos City, Philippines, Cueto, Uy, and Diaz [18] evaluated the carbon dioxide (CO₂) savings by transitioning from motorised tricycles to e-tricycles and discovered that the average daily CO₂ output of a motorised tricycle was 20.98 kg per unit per day. Saenz, Figliozzi, and Faulin [19] examined the carbon footprint of an e-tricycle logistics business against a diesel-powered urban logistics company. They found that total carbon dioxide equivalent (CO₂e) emissions are reduced by 51 to 72%. Temporelli et al. [20] assessed an e-tricycle called SUM-X produced by an Italian society, which was used by a logistics operator for last-mile deliveries in Padua city centre. The assessed e-tricycle has a load capacity of about 250 kg and is equipped with a battery with 60 km of autonomy and was compared with two vans, namely Nissan e-NV200 (Nissan Motor Co., Ltd., Japan) and the Ford Transit Connect (Ford Motor Company, Dearborn, Michigan, U.S.A.). According to the data, replacing the diesel van with an e-tricycle reduces last-mile logistics carbon emissions by around 250 g CO₂e/km. Gumasing et al. [21] created a service quality model for sustainable e-tricycle operations in the city of Manila, Philippines, to quantify the impact of 10 sustainability factors. Even though e-tricycle could save great energy, most of the existing studies are focused on passenger mobility. The performance of e-tricycles used in urban cargo deliveries is rarely studied [22]. The rapid development of emerging countries [23] indicates that understanding their urbanisation and improving their business models is crucial. In that light, assessing the performance of e-tricycles is important for their widespread adoption in urban cargo deliveries.

The main goal of this study is to assess the energy, economic, and environmental performance of e-tricycles used on medium-size cargo deliveries in Dar es Salaam city, Tanzania. The contributions of this study are as follows:

- The e-tricycle is designed based on the medium-sized cargo routed in Dar es Salaam city. The main target is to quantify the energy consumption in specific routes within the city.
- The economic performance of the e-tricycle based on the medium-size cargo and routes is assessed.
- The energy, economic, and environmental benefits of the e-tricycle are compared with traditional modes such as motorcycles, tuk-tuks, and light-duty cars.

2 | Materials and Methods

2.1 | Existing Modes of Urban Deliveries in the Study Area

Tanzania, one of the emerging economies, is still having trouble putting efficient urban delivery systems in place [24]. The main reasons for transportation problems, which lead to subpar urban delivery and higher service costs, are centralised services and traffic congestion [25]. Dar es Salaam has the highest percentage of cars [26], population, and per capita income in comparison to other regions in the country. This is because Dar es Salaam is a major port in Tanzania and an industrial hub. Additionally, Dar es Salaam dominates the economic and industrial sectors [27]. In the fastest growing cities, like Dar es Salaam, Tanzania, there is a

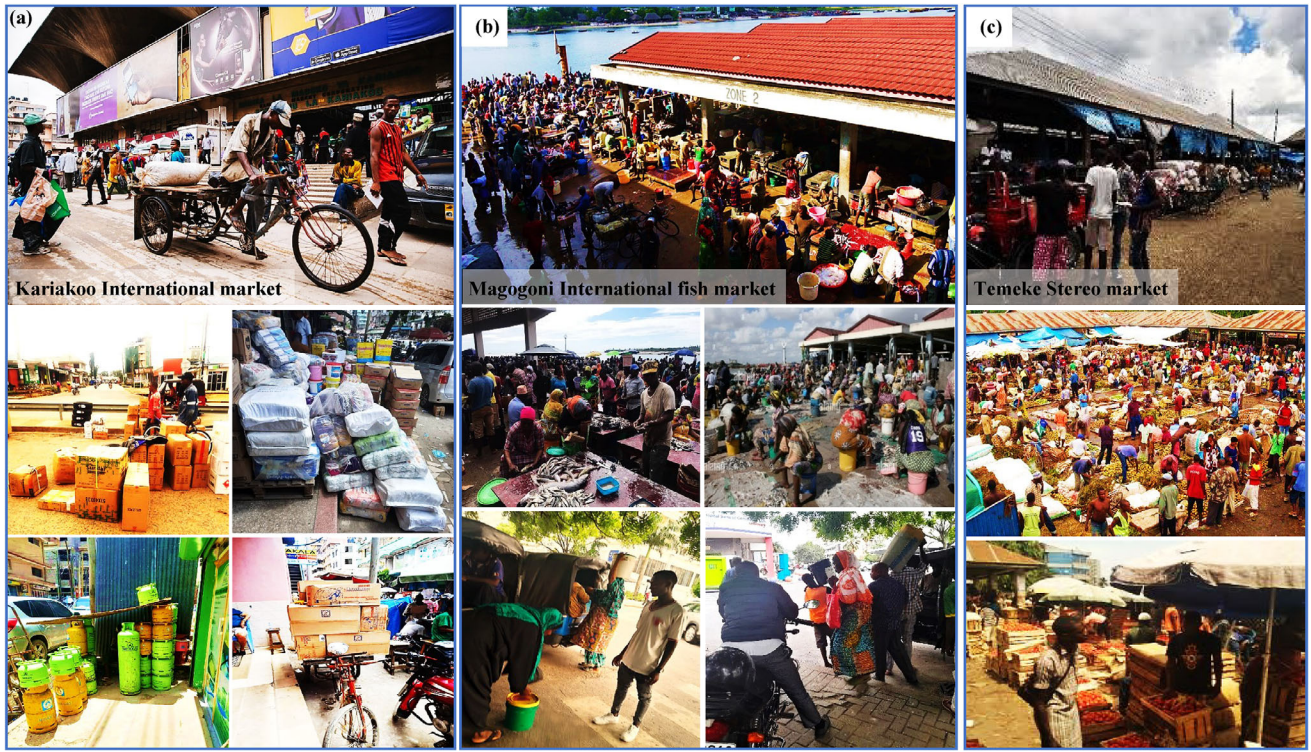


FIGURE 1 | Some of the busiest and most popular locations that generate plentiful medium-sized goods in Dar es Salaam (a) Kariakoo International Market, (b) Magogoni International Fish Market, and (c) Temeke Stereo Market.

sharp rise in population growth, which in turn raises the demand for mobility particularly good deliveries. Urban mobility in Dar es Salaam has long been described as a difficult task, requiring city dwellers to devote numerous hours to transporting commodities to the areas where low-income residents live. For instance, East and Central Africa's largest and busiest food market, Kariakoo International Market (Figure 1a), offers top-notch, sustainable services for food products, agricultural supplies, and merchandise (see small pictures in Figure 1a). The market serves millions of people in East and Central Africa, and most medium-sized goods are delivered to locals using the traditional modes of transport such as public buses, minibuses called daladala, minivans, tuk-tuks, traditional tricycles, and motorcycles (see in Figure 2). Because of long distances, goods suppliers are prone to employ other modes of transportation for last-mile connectivity, which add costs to the consumers. Another example is Magogoni International Fish Market (Figure 1b). This is the busiest market and a very popular supplier of seafood varieties to fishermen and consumers. This place generates plentiful medium-sized goods (small pictures in Figure 1b) which are distributed throughout the city via traditional modes of transportation (Figure 2). Temeke Stereo Market is another busy place that distributes a variety of medium-sized packaged vegetables and fruits throughout the city. The majority of these renowned locations are congested and positioned far from residential areas. As a result, goods vendors may use traditional modes of transportation and connect to alternative transportation providers for last-mile connection, which increases delivery time and service costs. For last-mile connectivity, motorcycles and tuk-tuks have been a common mode of goods delivery. Nonetheless, their operation is a substantial source of road emissions. In general, Figure 1 shows some of the operations in international marketplaces that are accessible in

Dar es Salaam and act as hubs for neighbouring countries, while Figure 2 shows the typical delivery methods that may need to give way to e-tricycles.

2.2 | Energy Modelling

The proposed e-tricycle model is shown in Figure 3. For the e-tricycle to move, it should overcome all the resisting forces such as aerodynamic force (F_{aero}), inertial force (F_g), and frictional force (F_{RR}) as indicated in Figure 3. F_{aero} accounts for the resistance due to the wind (see in Equation 1), F_g for the total vehicle weight (see in Equation 2), and F_{RR} for the friction between wheels and road (see in Equation 3). The tractive force (F_{tr}) which is a resultant of the three resisting forces is computed as shown in Equation (4). The resultant mechanical power (P) and torque (T) of the drivetrain needed to propel the e-tricycle is given in Equations (5) and (6) [28], respectively.

$$F_{aero} = \frac{1}{2} A_f C_D \rho_{air} (V_d - V_w)^2 \quad (1)$$

$$F_g = (m_d + m_p) g \sin(\theta) = Mg \sin(\theta) \quad (2)$$

$$F_{RR} = (m_d + m_p) g C_{RR} \cos(\theta) = Mg C_{RR} \cos(\theta) \quad (3)$$

$$F_{tr} = F_{aero} + F_g + F_{RR} \quad (4)$$

$$P = \frac{F_{tr} \times V_d}{\eta} \quad (5)$$

$$T = \frac{P}{\omega} = \frac{60 \times P}{2\pi N} \quad (6)$$



FIGURE 2 | Traditional modes of delivering commodities to locals (a) public buses (b) light-duty cars (c) tuk-tuks (d) motorcycles (e) traditional tricycles.

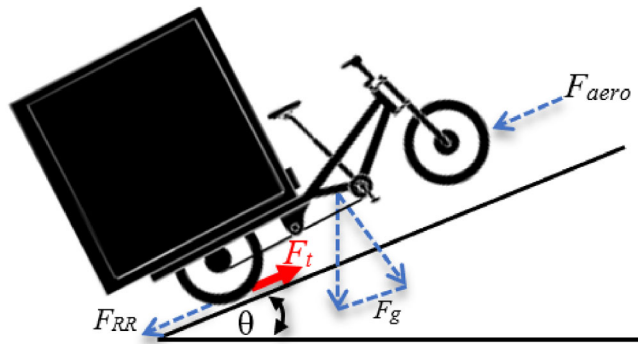


FIGURE 3 | Resisting forces acting on an e-tricycle.

where A_f is the front area of the e-tricycle (m^2), C_D is the drag coefficient due to air, ρ is the density of air (kg/m^3), V_d is the longitudinal velocity of the e-tricycle (m/s), V_w is the mean wind velocity (m/s), M is the sum of the dead weight and payload weight of the e-tricycle (kg), g is the force of gravity (m/s^2), θ is the road angle (degrees), C_{RR} is the frictional coefficient between the tires of the e-tricycle and road, η is the motor efficiency (%), and N is the maximum powertrain speed (rpm).

3 | Energy Performance of the E-Tricycle

For e-tricycle, the energy performance is evaluated by four parameters, namely, energy consumption per one-way trip, energy consumption per round trip, number of trips per charge, and

mileage per unit. After computing the propelling power (P), then the energy consumed per kilometre (E_{cons}), usable energy required to cover only the distance of route i (E_{usable}), and total energy required to cover the distance of route i plus the small portion of energy remaining in the battery of the e-tricycle (E_{total}) can be computed using Equations (7)–(9) [28]. The maximum values used in the computation of power required to propel the e-tricycle are shown in Table 1.

$$E_{cons} = \frac{P}{V_d} \quad (7)$$

$$E_{usable} = E_{cons} \times d_{trip} \quad (8)$$

$$E_{total} = E_{usable} \times DoD \quad (9)$$

where E_{cons} is the energy consumption of the motor (kWh/km), E_{usable} is the usable energy required to cover only the distance of route i , (kWh), d_{trip} is the trip distance (km), E_{total} is the total energy required for plying in a route i plus the energy remaining in the storage system to account for its safety (kWh), and DoD is the depth of discharge of the energy storage system (typically 20% state of charge or 1.25).

To estimate the E_{usable} and later E_{total} , the study is being undertaken in some selected routes in Gongo la Mboto and Mbagala rangi tatu areas, which are located in Ilala and Temeke Municipalities, respectively, to analyse energy consumption for urban delivery. The reason for selecting Gongo la Mboto and Mbagala for this study is the abundance of residential areas and passengers travelling to the city centre, shopping malls, offices,

TABLE 1 | Maximum values of system parameters.

Parameters	Abbreviation	Unit	Values	Ref
Dead weight	m_d	kg	60.00 ^a	Greyson et al. [28]
Payload weight	m_p	kg	250.0 ^b	Temporelli et al. [20]
Total weight	M	kg	310.0 ^c	This study
Road angle	θ	°	3.40	Greyson et al. [29]
Air density	ρ	kg/m ³	1.29	Greyson et al. [28]
Drag coefficient	C_D	—	0.60	Greyson et al. [28]
Frontal area	A_f	m ²	2.20	This study
Gravitational force	g	m/s ²	9.81	Greyson et al. [28]
Driving velocity	V_d	km/h	50.0	This study
Coefficient of friction	C_R	—	0.48	Greyson et al. [28]
Motor efficiency	η	%	0.83	Temporelli et al. [20]

^aThe dry weight of a motorcycle without an engine but including an electric motor and battery.

^bAn average weight of two passengers and a driver, each having 80 kg.

^cA sum of dead weight and payload weight.

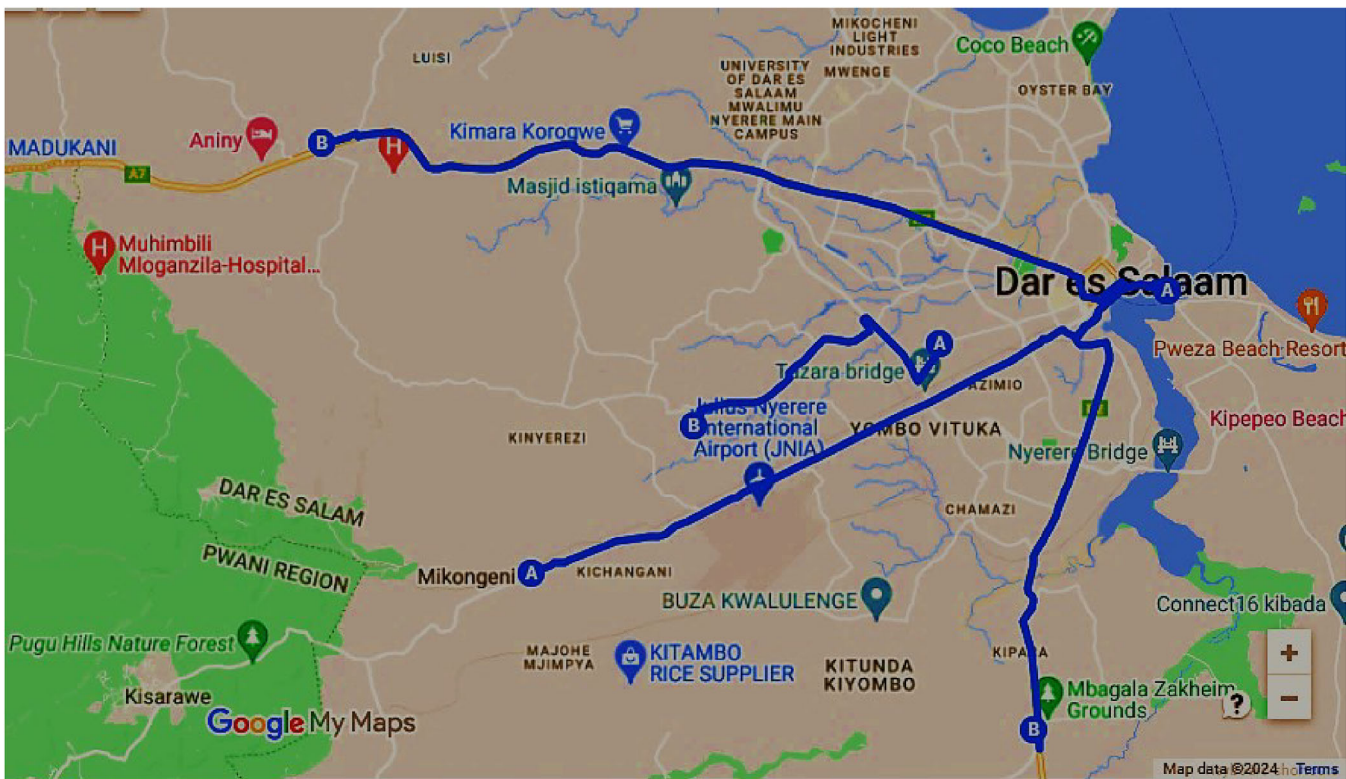


FIGURE 4 | The selected routes under the study area of Dar es Salaam city (Source: Google Map).

schools, markets, and regional bus terminals via interconnection with city buses, bus rapid transit, metropolitan railway, and regional bus services. The energy requirement for urban delivery in the targeted area is estimated from Gongo la Mboto bus stop, Mbezi Magufuli bus stop, and Mbagala bus terminal to Kivukoni Fish Market, Tegeta Nyuki terminal, and Tabata Segerea bus terminal. The referred bus stops can be seen on Google Maps as presented in Figure 4, and their average mileages are indicated in Table 2.

4 | Economic Modelling

The economic assessment involves the computation of cost per km (C_{km}), cost per trip (C_{trip}), and cost per kg of luggage (C_{kg}). The cost per km (C_{km}) comprises of operating cost (C_{op}) (typically fuel cost), maintenance cost (C_m), and trip distance (d_{trip}) as shown in Equation (10) [30]. The C_{km} describes the cost for delivery per kilometre and is expressed in TZS/km. Another important parameter is cost per trip (C_{trip}) which expresses the cost of

TABLE 2 | One-way average distance of routes in the targeted area.

Route description	Route name	One-way distance (km)
Kivukoni Fish Market to Tegeta Nyuki terminal	Route 1	25
Kivukoni Fish Market to Mbezi Magufuli bus terminal	Route 2	23
Buguruni Rozana to Tabata Segerea bus terminal	Route 3	10
Gongo la Mboto to Kivukoni Fish Market	Route 4	18
Kivukoni Fish Market to Mbagala Rangi Tatu	Route 5	15

delivery on a specified route i . The C_{trip} is shown in Equation (11) [31] and expressed in TZS.

$$C_{km} = \frac{C_{op} + C_m}{d_{trip}} \quad (10)$$

$$C_{trip} = C_{km} \times d_{trip} \quad (11)$$

$$C_{kg} = \frac{C_{op} + C_m}{m_p} \quad (12)$$

where C_{km} is the cost per kilometre (TZS/km), C_{op} is the operating cost (TZS), C_m is the maintenance cost (TZS), and C_{trip} is the trip cost (TZS).

5 | Environmental Saving

Since the e-tricycle uses batteries as a source of energy, no tank-to-wheel emission is generated during on-road driving. Therefore, the environmental impact would account only for the emissions from traditional modes only. The most studied parameter, carbon dioxide (CO_2), will be assessed based on the amount that could be generated from the liquid fuel. Since the liquid fuel used in all traditional modes under study is the same (petrol), then the emission intensity (e_{in}) of 2.162 kg of CO_2 per litre of petrol is used [32]. The fuel consumption for each mode was taken as an average fuel consumed per kilometre. This consumption was obtained from the local drivers of each traditional mode. Then, fuel consumption was mapped on the routes to obtain the total fuel to be consumed for the specific routes. After obtaining the total fuel consumption for specific routes and the emission intensity, the total emissions were computed. For this study, emission per day (e_{day}), emission per trip (e_{trip}), and emission per luggage ($e_{luggage}$) were estimated based on Equations (12) to (14) [28].

$$e_{day} = e_{cons} \times e_{int} \quad (12)$$

$$e_{trip} = \frac{e_{day}}{\#trips} \quad (13)$$

$$e_{luggage} = \frac{e_{day}}{m_p} \quad (14)$$

where e_{day} is the emission per day (kg CO_{2e} /day), e_{in} is the emission intensity of a specific transport mode (kg CO_{2e} /L), and $e_{luggage}$ is the emission per luggage (kg CO_{2e} /kg of luggage).

6 | Comparison Between E-Tricycle and Traditional Modes of Deliveries

The traditional modes of deliveries used in the study are motorcycle, tuk-tuk, and light-duty car (LDC). The properties of these traditional modes of deliveries are shown in Table A1 (see in the Appendix). The comparison is made on the energy, economy, and environmental benefits by computing the difference between e-tricycle and its counterpart. For this case, the energy content of petrol fuel used in a traditional mode is converted to its electricity equivalent (8.9 kWh/L) before calculation. Then, the total energy of traditional modes given in kWh (see in Table 3) was compared with that of e-tricycle. The obtained difference is taken as the energy that could be saved by using an e-tricycle.

For the proper estimation, the following assumptions are used:

Assumptions

- Speed-dependent rolling resistance is assumed to be fixed at all vehicle speeds,
- Wind speed is assumed to be constant,
- E-tricycle battery capacity: 5 kWh (locally available battery capacity of electric tuk-tuks),
- Fuel consumption: 0.021 kWh/km for e-tricycle, 5 L/day for motorcycle, 8 L/day for tuk-tuk, and 20 L/day for LDC,
- Daily mileages: 120 km/day for motorcycle, 100 km/day for tuk-tuk, and 200 km/day for LDC,
- Energy conversion factor: 8.9 kWh/L of petrol [33],
- Energy consumption: (5L/day÷120 km/day × 8.9 kWh/L) 0.37 kWh/km for motorcycle, (8L/day÷100 km/day × 8.9 kWh/L) 0.712 kWh/km for tuk-tuk, and (20L/day÷200 km/day × 8.9 kWh/L) 0.89 kWh/km for LDC,
- Fuel prices: TZS 354/kWh of electricity [34], TZS 3011/L of petrol (as of October 2024) [35],
- Operating cost: TZS 1770 for e-tricycle, TZS 16,000 for motorcycle, TZS 25,600 for tuk-tuk, and TZS 64,000 for LDC,
- Maintenance cost: TZS 10,000 for e-tricycle, motorcycle, and tuk-tuk; TZS 20,000 for LDC,
- Emission intensity: 2.162 kg CO_{2e} /L of petrol [32].

The computed energy consumption for the e-tricycle, motorcycle, tuk-tuk, and LDC is shown in Table 3. The consumption is based

TABLE 3 | Energy consumption in e-tricycle, motorcycle, tuk-tuk, and light-duty car (LDC).

Routes	Mileage (km)	e-tricycle	Motorcycle	Tuk-tuk	Light-duty car (LDC)
Route 1	25	0.536	9.271	17.800	22.250
Route 2	23	0.493	8.529	16.376	20.470
Route 3	10	0.214	3.708	7.120	8.900
Route 4	18	0.386	6.675	12.816	16.020
Route 5	15	0.321	5.563	10.680	13.350

TABLE 4 | Computation of CO_{2e} emissions from the motorcycle, tuk-tuk, and light-duty car (LDC).

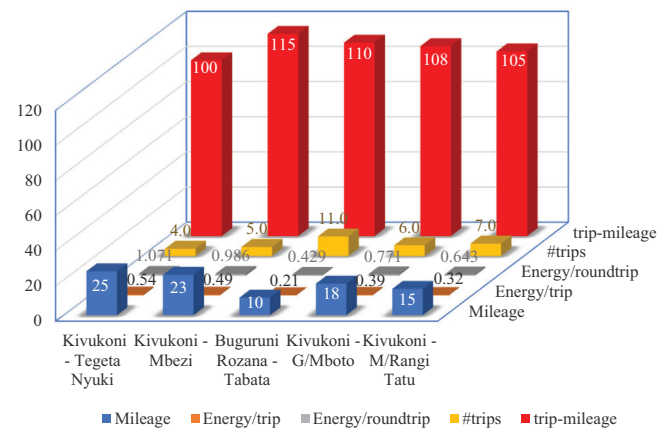
Particulars	Motorcycle	Tuk-tuk	Light-duty car (LDC)
Daily fuel use (L/day)	5	8	20
Daily mileage (km/day)	120	100	200
Daily trips	4	4	4
Luggage weight (kg)	250	250	250
Fuel consumption (L/km)	$(5 \div 120) = 0.042$	$(8 \div 100) = 0.080$	$(20 \div 200) = 0.100$
Emissions per day (kgCO _{2e} /day)	$(5 \times 2.162) = 10.810$	$(8 \times 2.162) = 17.296$	$(20 \times 2.162) = 43.240$
Emissions per trip (kgCO _{2e} /trip)	$(10.810 \div 4) = 2.702$	$(17.296 \div 4) = 4.324$	$(43.240 \div 4) = 10.810$
Emissions per luggage (kgCO _{2e} /luggage)	$(10.810 \div 250) = 0.04324$	$(17.296 \div 250) = 0.06918$	$(43.240 \div 250) = 0.17296$

on the route mileages. These values were used to estimate the energy saving per trip and energy saving per kilometre. The CO_{2e} emissions of the motorcycle, tuk-tuk, and LDC are computed as shown in Table 4. The values are used to highlight the CO_{2e} emissions per day, CO_{2e} emissions per trip, and CO_{2e} emissions per luggage.

7 | Results and Discussion

7.1 | Energy Performance

From Equation (5), the computed tractive power (P) required to carry the total load of 250 kg is about 525.78 W. In this work, a three-phase BLDC motor known as a rear wheel hub motor with a standard value of 750 W is chosen. Figure 5 shows the energy performance of the e-tricycle in terms of mileage, energy per one-way trip, energy per roundtrip, number of trips per charge, and trip mileage when operated in five different routes. In Figure 5, the blue bars represent the one-way trip mileages which are also given in Table 2. The orange bars represent the energy per one-way trip. It can be seen that energy consumed per trip ranges between 0.21 to 0.54 kWh/trip, with the shortest trip mileage consuming the lowest energy. When considering the roundtrip distance (grey bars), from the source point to the destination and back to the source, the energy consumption ranges between 0.429 to 1.071 kWh/roundtrip. The yellow bars show the number of trips per charge (#trips). By considering the average battery pack capacity of 5 kWh, the number of roundtrips per charge could be between 4 and 11, varying with the route length. With these number of trips, the 5 kWh energy storage of the e-tricycle could reach 100 and 115 km trip-mileage daily on a single charge (see the red bars).

**FIGURE 5** | Comparison of energy consumption of the e-tricycle in different routes.

The comparison of energy consumption between the e-tricycle and traditional modes of urban deliveries (motorcycle, tuk-tuk, and LDC) on different routes is shown in Figure 6. The x-axis of Figure 6 shows the routes, while the z-axis shows different modes under comparison. The energy consumption of different delivery modes is shown in the y-axis. It can be seen that the e-tricycle consumes between 0.429 and 1.071 kWh/km, while the light-duty car (LDC) consumes between 17.8 and 44.5 kWh/km per round trip. Other modes, such as motorcycles and tuk-tuks consume between 7.417 and 18.542 and 14.240 and 35.6 kWh/km. All modes consumed the most in the longest route and least in the shortest route. By considering the longest route with 25 km (see Table 2), this means that the e-tricycle could save energy around 17.469 kWh/km against the motorcycle, equivalent to 94.22%, which indicates a huge potential for energy saving when adopting

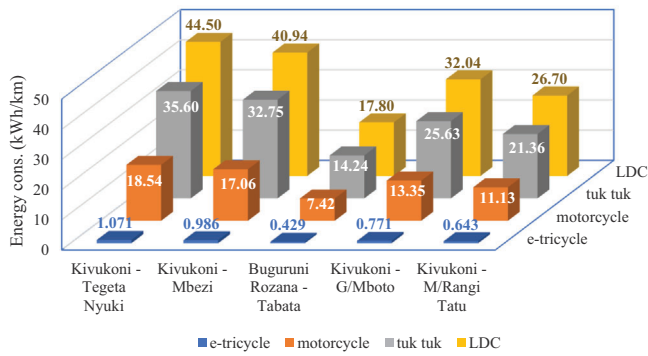


FIGURE 6 | Energy consumption of an e-tricycle and other traditional modes of urban deliveries.

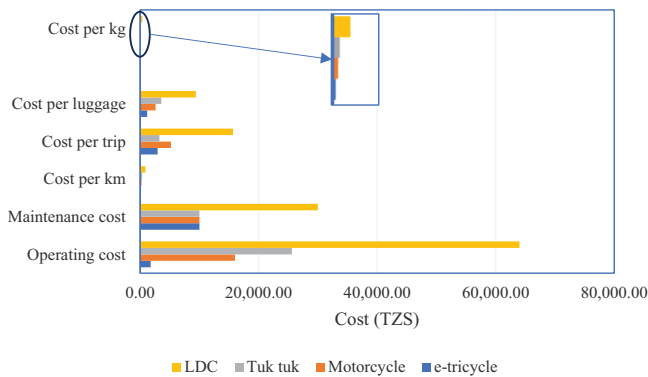


FIGURE 7 | Economic performance of e-tricycle and other delivery modes.

e-tricycles on a large scale. When compared to the LDC on the same route, the e-tricycle could reduce energy consumption by 43.429 kWh/km or 97.59%. In the shortest route of 15 km, the e-tricycle could save energy around 10.417 kWh/km against the motorcycle, equivalent to 94.22%. On the other side, when compared with the LDC, the energy saving in e-tricycle reaches 97.59%. Overall, the energy-saving potential of the e-tricycle ranges between 94.22 to 97.59%.

7.2 | Economic Saving

Figure 7 shows the economic performance of delivering with the e-tricycle and other traditional delivery modes. Here, the comparison is made in the operating cost (fuel cost) and maintenance cost. Later, the running cost was translated per km, per trip, per luggage, and per kg of luggage. The fuel cost of petrol was considered as TZS 3,200 per litre while that of electricity was considered as TZS 354 per kWh. As referred to in Figure 7, it is revealed that the e-tricycle provides a significant reduction in operating costs (fuel cost) because its powertrain is fuelled by electricity, which is less expensive than liquid fuel. For the case of maintenance, all vehicles were regarded to use TZS 10,000 per month except the LDC, which incurred about TZS 30,000 per month. The total running cost was taken as the total of operating and maintenance costs. From Figure 7, it can be seen that operating cost takes the largest part in running cost. Moreover, the LDC had the highest running expenses, while the e-tricycle

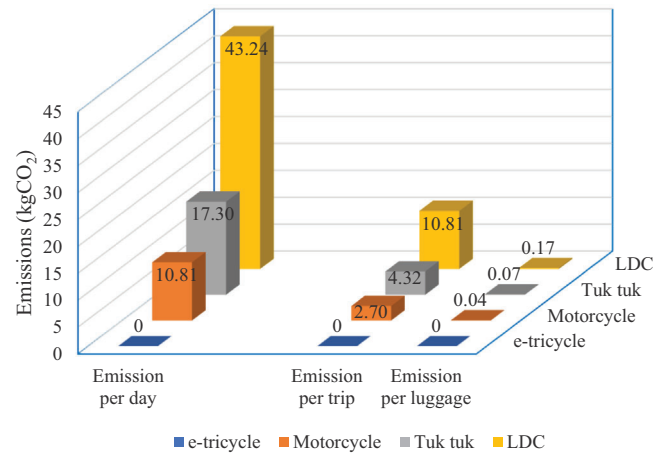


FIGURE 8 | Environmental performance of e-tricycle and other delivery modes.

had the lowest. When translated into the service cost, it can be seen that for e-tricycle, motorcycles, tuk-tuks, and LDCs, the cost per km is around TZS 120, 220, 320, and 870, respectively. In terms of cost per trip, the e-tricycle, motorcycle, tuk-tuk, and LDC cost around TZS 2,950, 5,200, 3,250, and 15,700. If the delivery is restricted to 250 kg in all modes and assuming that the maximum weight of luggage is 25 kg (10 pieces of luggage for each mode), the cost per luggage for the e-tricycle, motorcycle, tuk-tuk, and LDC could reach TZS 1,200, 2,600, 3,560, and 9,400, respectively. However, in most cases the number of luggage is uncontrollable. Instead, the weight could be the best parameter for control. If only 250 kg is to be delivered, the cost per kg for the e-tricycle, motorcycle, tuk-tuk, and LDC could reach around TZS 50, 110, 150, and 380.

From the findings, it is revealed that e-tricycle can save around TZS 750 per km, equivalent to 86% when compared to LDC. This means the e-tricycle has the potential to reduce about 86% of the travelling cost per km as compared to LDC. On the other hand, when compared to motorcycles and tuk-tuks, the e-tricycle shows savings of approximately TZS 100 and 200 per km or 45.5% and 62.5%, respectively. In terms of cost per trip, the e-tricycle offers the lowest cost, about 5.3, 1.1, and 1.8 times cheaper compared to LDC, tuk-tuks, and motorcycles, respectively. Comparatively, by restricting the deliveries to 250 kg, the e-tricycle could have the lowest cost per kg, about 7.6, 3.0, and 2.2 times cheaper compared to LDC, tuk-tuks, and motorcycles. All the studied economic indicators favours the e-tricycle in urban deliveries, indicating a huge potential to benefit e-tricycle users. The e-tricycle also presents a chance to take advantage of the subsidies and incentives, as it is in line with the global trend towards cleaner mobility. The usage of electricity paves the way for the use of renewable energy, which drastically lowers carbon emissions. This helps to improve air quality, mitigate climate change, and create a more sustainable transportation system while reducing our need for fossil fuels.

7.3 | Environmental Saving

Figure 8 shows the emission performance of the e-tricycle and other modes of delivery. The x-axis of Figure 8 shows the emission

per day, emission per trip, and emission per luggage, while the z-axis shows the delivery modes. It can be seen that the e-tricycle does not produce road emissions due to its electric propulsion system. However, a significant amount of emission can be seen in its counterparts. It can be seen that motorcycles, tuk-tuks, and LDC emit between 10.81 and 43.24 kgCO₂ per day. These levels are due to their daily fuel consumption at an emission intensity of 2.262 kgCO₂ per litre of petrol. By considering that the designed e-tricycle can make around 4 trips per day with a single charge, the emission per trip in all modes can be estimated to range between 2.70 to 10.81 kgCO₂ per trip. Since each mode is regarded to carry around 250 kg, the emission per kg of luggage could range from 0.04 to 0.17 kgCO₂ per luggage weight. In all cases, the LDC had the highest emission intensity, followed by the tuk-tuk and motorcycle.

8 | Conclusion

The study was designed to understand the performance of the e-tricycle in comparison with the traditional delivery modes such as petrol-powered motorcycle, tuk-tuk, and light-duty car (LDC) in the targeted routes. The e-tricycle consumes between 0.21 to 0.54 kWh/one-way trip, with the shortest trip mileage consuming the lowest energy. For the round trip, the e-tricycle consumes approximately 0.429 to 1.071 kWh/round trip. With the average battery pack capacity of 5 kWh, the number of round trips per charge could be between 4 and 11, depending on the route length. With this number of trips, the 5 kWh energy storage of the e-tricycle could reach 100 and 115 km trip-mileage daily on a single charge. The e-tricycle consumes between 0.429 and 1.071 kWh/km, while the light-duty car (LDC) consumes between 17.8 and 44.5 kWh/km per round trip. Other modes, such as motorcycle and tuk-tuk, consume between 7.417 and 18.542; and 14.240 and 35.6 kWh/km, depending on the route length. In terms of economy, e-tricycle, motorcycle, tuk-tuk, and LDC could reach around TZS 120, 220, 320, and 870 per km. In terms of cost per trip, they could cost around TZS 2,950, 5,200, 3,250, and 15,700. When restricting to the delivery weight of 250 kg, the cost per kg for the e-tricycle, motorcycle, tuk-tuk, and LDC could reach around TZS 50, 110, 150, and 380. The e-tricycle provides a clean delivery. In comparison with its counterparts, motorcycle, tuk-tuk, and LDC emit between 10.81 and 43.24 kgCO₂ per day, or between 2.70 to 10.81 kgCO₂ per trip.

9 | Recommendation

The traditional modes of delivery involved in this study are all powered by fossil fuels. Consequently, the delivery cost surges as a result of fuel and maintenance costs. Road emissions are also produced as a byproduct of fossil fuel burning in these traditional modes of delivery. Currently, the deployment of e-tricycle for urban delivery has not received much attention, probably due to their less adoption and little awareness of their benefits. This study recommends executing the road test to acquire the real-world energy consumption of the e-tricycle. The data could be key in the future advancement of the vehicle, establishment, and distribution of the charging stations. The implications (e.g., electricity bills, emission cutdown in electricity production, and time-saving) of using renewable energy resources [36] for charging

e-tricycles should be further investigated. Further studies should highlight the economics of various battery charging methods for promoting the e-tricycle business. When more e-tricycles and charging station models become operational, more studies should focus on identifying the optimal allocation of charging stations within the city and the viability of allocating charging stations in key areas such as shopping centres, international markets, public and rapid bus corridors, etc. Operational factors that impact the e-tricycle operation (trip length, drivers' age, driving behaviours, loading, road types, and climatic conditions) should be well understood and addressed. Attention should also be paid to raising awareness among e-tricycle users regarding the benefits of reducing road emissions. The absence of an EV policy and local manufacturing technology restricts the deployment of e-tricycle for delivery in the Tanzanian context, which must be resolved. To further address the remaining barriers to EV adoption, policymakers should strengthen incentives in the form of carbon credits for e-tricycle and charging station owners. The gain of this credit could be translated from the annual emission that could be avoided by using e-tricycles or charging using renewable energy. This credit could be indirectly advantageous to the beneficiaries by reducing their operational costs, such as municipal parking fees, road licences, etc. As the country is drafting the national EV policy framework, the e-tricycle could be included in the tax incentives. The key stakeholder, the Tanzania E-Mobility Association (TAEMA), could recommend the government move to complete knockdown (CKD) imports and allow companies to access the East African countries (EAC) CKD rates to reduce taxation by a third and make e-tricycle cost-competitive.

Nomenclature

A_f	Frontal area (m ²)
CD	Drag coefficient due to air (dimensionless)
C_{kg}	Cost per luggage (TZS/kg)
C_{km}	Cost per kilometre (TZS/km)
C_m	Cost of maintenance (TZS)
CO_{2e}	Carbon dioxide equivalent
C_{op}	Cost of operation (TZS)
CRR	Frictional coefficient between tires and road
C_{trip}	Cost per trip (TZS/trip)
d	Distance (km)
DoD	Depth of discharge (%)
E_{cons}	Energy consumed per kilometre (kWh/km)
e_{day}	Emissions per day (kgCO _{2e} /day)
$e_{luggage}$	Emissions per luggage (kgCO _{2e} /kg of luggage)
E_{total}	Total energy required for route i (kWh)
e_{trip}	Emissions per trip (kgCO _{2e} /trip)
E_{usable}	Usable energy required for route i (kWh)
F_{aero}	Aerodynamic force (N)
F_g	Force of gravity (N)
FRR	Frictional force (N)

g	Force of gravity (m/s^2)
M	Mass of dead weight and payload weight (kg)
md	Dead weight (kg)
mp	Payload weight (kg)
N	Maximum powertrain speed (rpm)
P	Propelling power (/kW)
Vd	Longitudinal velocity of the vehicle (m/s)
Vw	Mean wind velocity (m/s)

Subscripts

i	Specific route
trip	Trip
cons	Consumption
usable	Usable
total	Total
op	Operational
m	Maintenance
km	Kilometre
kg	Kilogram
day	Day
<i>luggage</i>	Luggage

Greek Symbols

ρ	Density of air (kg/m^3)
θ	Road angle

Author Contributions

O.S.S. and P.V.C. initiated the concept: G.B.G., J.H.H., K.A.G., O.S.S., R.O.K., and P.V.C. prepared the modelling, collected validation data, analysed data, and wrote the manuscript.

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Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

Data will be made available on request.

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Appendix

TABLE A1 | Properties of the motorcycle, tuk-tuk, and light-duty car (LDC) used in this study.

Specifications	Motorcycle	Tuk-tuk	Light-duty car (LDC)
Engine type	Single cylinder 4-stroke	Two cylinders 4-stroke	Four cylinders 4-stroke
Displacement	125 cc	218.6 cc	1197 cc
Maximum power	2.2 kW at 350 rpm	7.7 kW at 6000 rpm	59.4 kW at 5750 rpm
Maximum torque	10 Nm at 300 rpm	17 Nm at 5500 rpm	104.4 Nm at 3000 rpm
Fuel	Petrol	Petrol	Petrol
Fuel economy	120 km/day	100 km/day	200 km/day
Fuel tank capacity	1 L/30 km	1 L / 25 km	1 L / 35 km
Gross weight	260 kg	790 kg	1600 kg
Kerb weight	150 kg	490 kg	860 kg
Payload	110 kg	300 kg	740 kg